

Lecture 7- Part II

The budget line

ECON101

Fall 2020



Budget Set and the budget line of the consumer

- Represent what is feasible to consume given market prices and income
- Two goods, x_1 , x_2 , and the consumer has income I , while facing prices p_1 and p_2 for goods 1 and 2.

For the consumer Total Expenditures $\leq$ Income

$$\underbrace{p_1 x_1}_{\substack{\text{amount} \\ \text{spent} \\ \text{on good 1}}} + \underbrace{p_2 x_2}_{\substack{\text{amount} \\ \text{spent} \\ \text{on good 2}}} \leq I \rightarrow \text{Budget Set}$$

any bundle $x = (x_1, x_2)$ that satisfies the inequality is the budget set.

When $p_1 x_1 + p_2 x_2 = I \rightarrow$ budget line and those bundles exhaust the income.

Graphically: $p_1 x_1 + p_2 x_2 \leq I$ EVA

budget line: $p_1 x_1 + p_2 x_2 = I$

↳ Solve for x_2 :

$$x_2 = \frac{I}{p_2} - \frac{p_1}{p_2} x_1 \quad \text{Eq. of Budget Line}$$

For bundle $A = (a_1, a_2)$

$$p_1 a_1 + p_2 a_2 < I$$

For bundle $B = (b_1, b_2)$

$$p_1 b_1 + p_2 b_2 = I$$

$C = (c_1, c_2)$

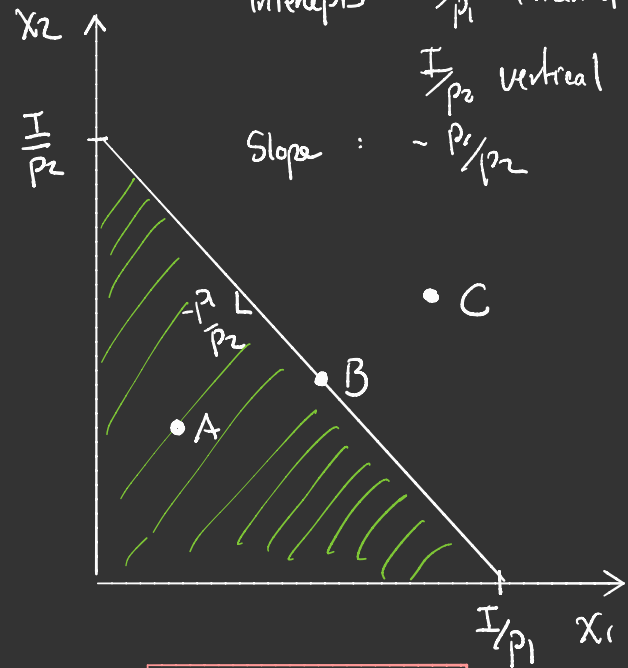
$$p_1 c_1 + p_2 c_2 > I$$

For the Budget line:

Intercepts: $\frac{I}{p_1}$ horizontal

$\frac{I}{p_2}$ vertical

Slope: $-\frac{p_1}{p_2}$



$\frac{p_1}{p_2}$: price ratio

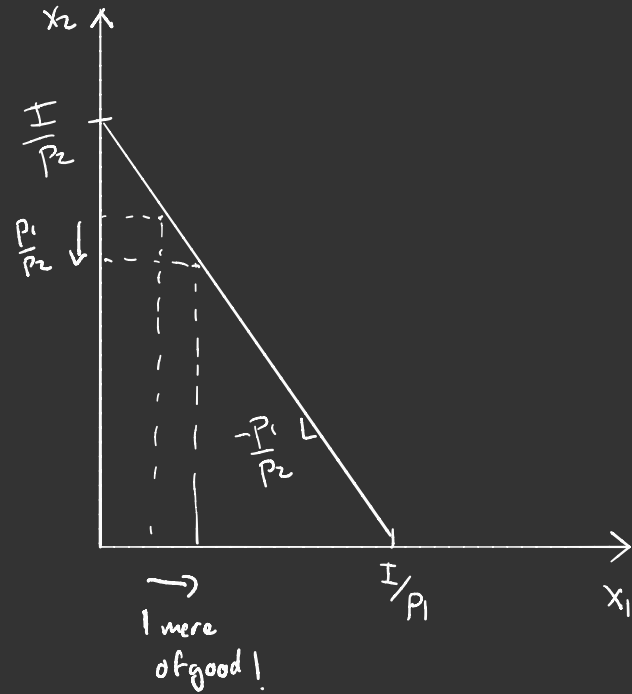
Slope of the budget line.

$$p_1 x_1 + p_2 x_2 = I, \quad \text{slope: } -\frac{p_1}{p_2}$$

$$\cdot \frac{p_1}{p_2} = \text{price ratio}$$

= market cost of opportunity
of good 1 in terms of
good 2

1 unit of good 1 costs you $\frac{p_1}{p_2}$ units
of good 2.



Changes in Income: parallel shift in the budget line

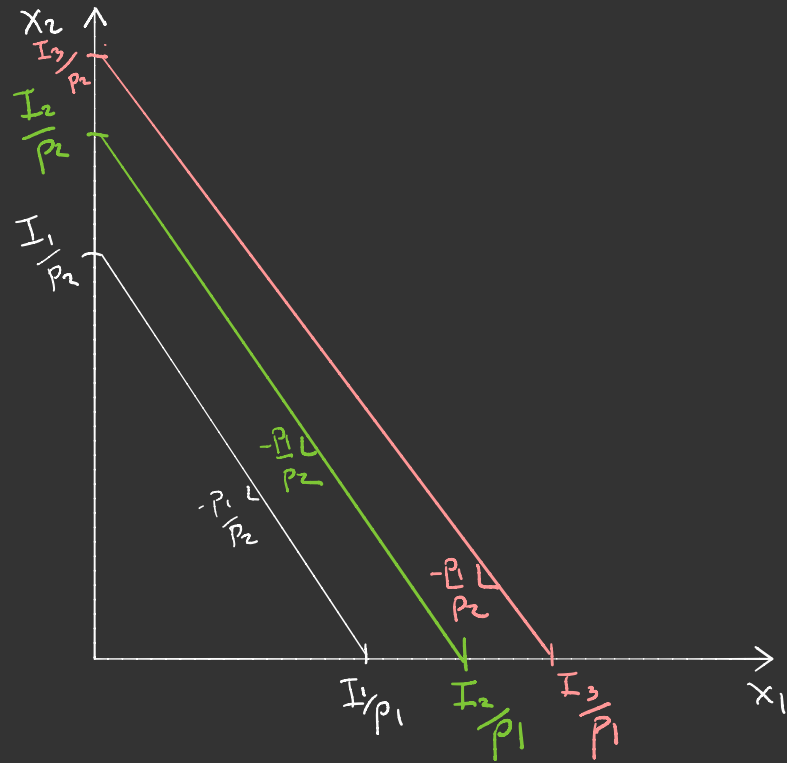
$$p_1 x_1 + p_2 x_2 = I$$

$$\hookrightarrow x_2 = \frac{I}{p_2} - \frac{p_1}{p_2} x_1$$

$$\text{Thus a } \Delta I = I_2 - I_1$$

$$I_1 < I_2 < I_3$$

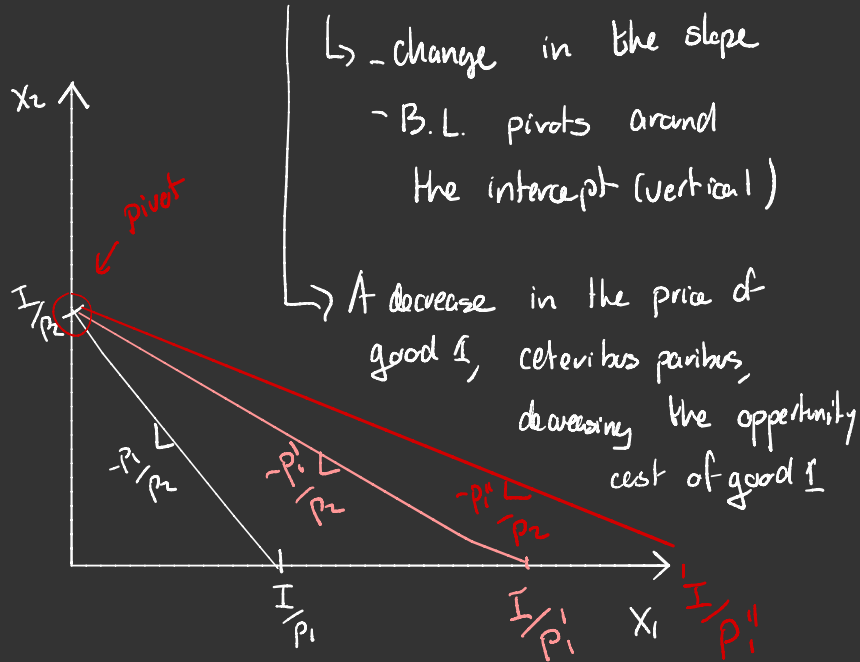
are parallel shifts in the budget line.



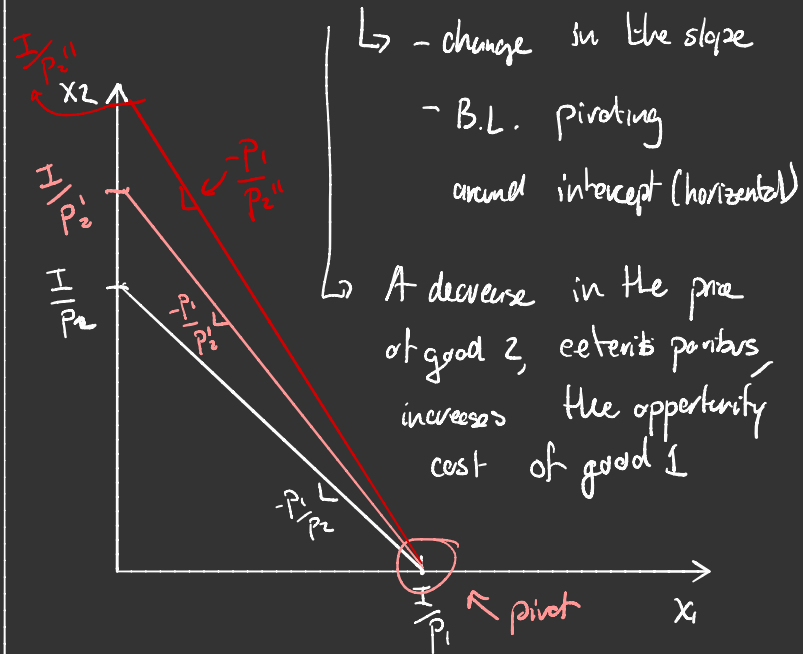
Changes in Prices: $p_1 x_1 + p_2 x_2 = I$

$$\text{eq.: } x_2 = \frac{I}{p_2} - \frac{p_1}{p_2} x_1$$

Change in $p_1 > p_1' > p_1''$



Change in $p_2 > p_2' > p_2''$



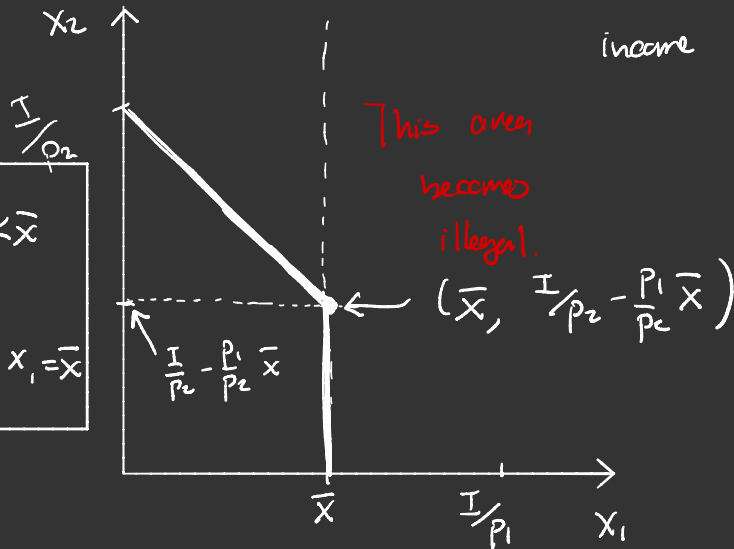
Non-Linear Budget Constraints

- Not all relevant B.C. are linear: $p_1 x_1 + p_2 x_2 = I$

→ Quantity restriction: suppose you are not allowed to buy more than $\bar{x}$ of good 1, prices of good 1 and 2 are still p_1 & p_2 , still have income I .

B.C. becomes a piece-wise function:

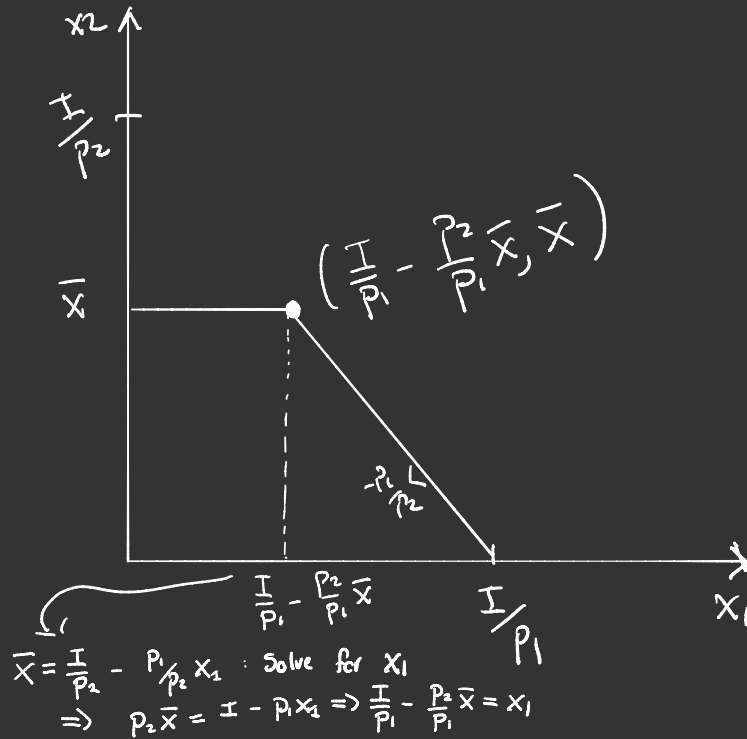
$$x_2 = \begin{cases} \frac{I}{p_2} - \frac{p_1}{p_2} x_1 & \text{if } x_1 < \bar{x} \\ [0, \frac{I}{p_2} - \frac{p_1}{p_2} \bar{x}] & \text{if } x_1 = \bar{x} \end{cases}$$



Quantity restriction on x_2 , by $\bar{x}$, same prices p_1 & p_2 and Income

The budget line becomes
piecewise :

$$x_2 = \begin{cases} \bar{x} & \text{for } 0 \leq x_1 \leq \frac{I}{p_1} - \frac{p_2}{p_1} \bar{x} \\ \frac{I}{p_2} - \frac{p_1}{p_2} x_1 & \text{for } \frac{I}{p_1} - \frac{p_2}{p_1} \bar{x} < x_1 \leq \frac{I}{p_1} \end{cases}$$



Vouchers, Coupons : piece of paper that allows you to buy a particular good
(promise)
in exchange of the coupon/voucher.

Example : Food Stamps.

- Popular income support program by governments.
- Used to support poor people (low incomes)
- Coupons are given that can be legally exchanged for
only food
- Popular with donors, and also given as agricultural subsidies
- Firms sometimes include food vouchers as part of the compensation.

- Suppose that $I = \$100$

- Price of food $P_F = \$1$

- Price of "all other goods" $P_G = \$1$

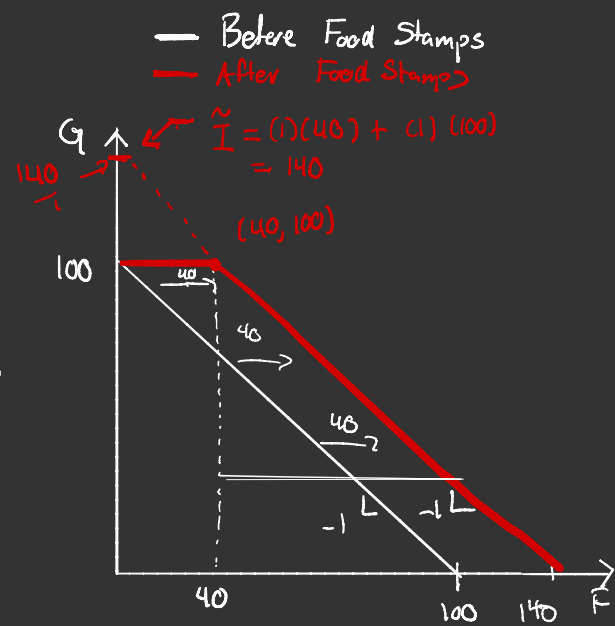
- We keep fixed the price of other goods at 1, i.e. it is normalized to 1, this is called "the numeraire".

- Budget Constraint (Line) is given $F + G = 100$

- You receive \$40 food stamps
↳ Key factor: Income available for other goods, remains the same, doesn't change with food stamps.

↳ Budget Set enlarges

↳ Budget Line has a kink.



Eq. of B.L. with food stamps

$$G = \begin{cases} 100 & 0 \leq F \leq 40 \\ 140 - F & 40 < F \leq 140 \end{cases}$$

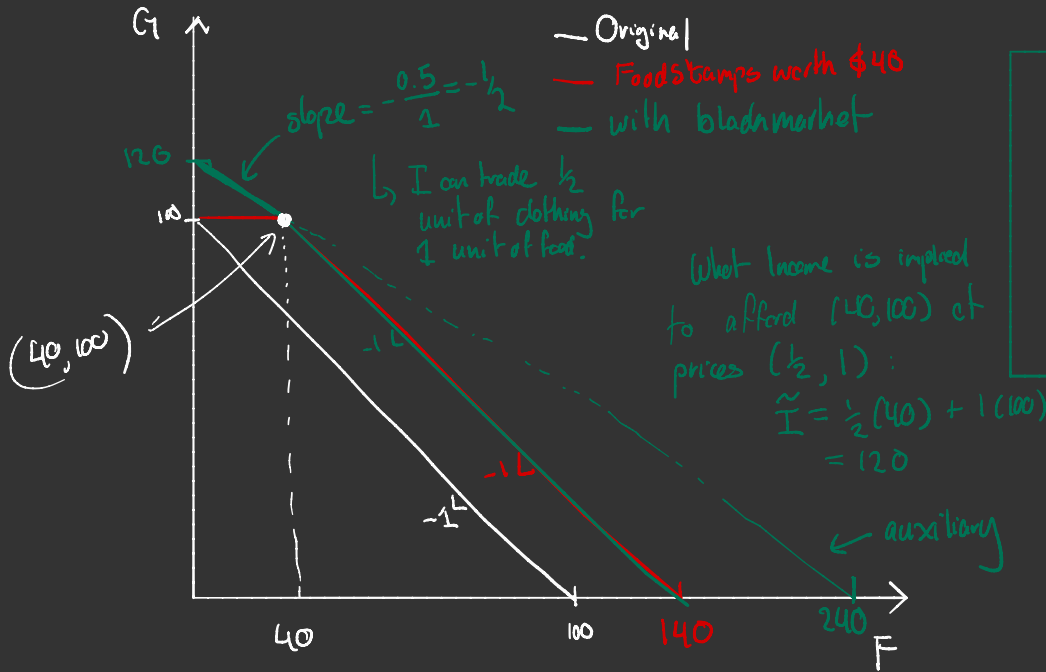
What if there is a Black Market for Food Stamps?

- I can say my food stamps for \$0.50 each?
- ↳ That is half what they are worth in food.

Black market Food Stamp

budget line:

- Expands
- Kinked (40, 100)



$$G = \begin{cases} 120 - \frac{1}{2}F & \text{if } 0 \leq F < 40 \\ 140 - F & \text{if } 40 \leq F \leq 140 \end{cases}$$

Eq. for B.L. with Food Stamps & Black Market

Quantity - Based Discount:

- Quantity - discount for large buyers.
- Suppose $p_2 = 1$, $I = 100$ constant
- $p_1 = 2$ for $0 \leq x_1 \leq 20 \Rightarrow -\frac{p_1}{p_2} = -2$
- $p_1 = 1$ for $x_1 > 20 \Rightarrow -\frac{p_1}{p_2} = -1$

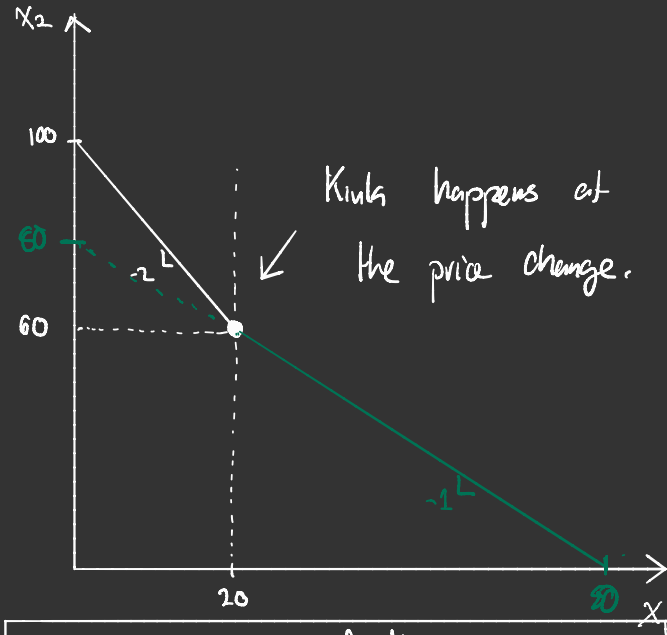
The eq. for — B.L. $x_2 = 100 - 2x_1$

The eq. for — B.L.

Find the implied income: the B.L. needs to pass through $(20, 60)$ at prices $(1, 1)$ we need an income

$$\tilde{I} = (1)(20) + (1)(60) = 80$$

$$x_2 = 80 - x_1$$



The joint eq. of this B.L. — + —
is
$$x_2 = \begin{cases} 100 - 2x_1 & \text{if } 0 \leq x_1 \leq 20 \\ 80 - x_1 & \text{if } 20 < x_1 \leq 80 \end{cases}$$